Technical Performance Catalog

Sandwich Panels with PUR, PIR, and Mineral Wool Cores

This technical catalog provides a detailed engineering and comparative analysis of the most advanced thermal insulation core materials on the market: Polyurethane (PUR), Polyisocyanurate (PIR), and Mineral Wool (MW). Specifically designed for engineers, architects, and project specifiers in high-performance industrial and commercial sectors.

International Standards Compliance

EN 13501-1 · EN ISO 6946 · FM Global 4880

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Global Engineering Standards · 2026

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1 Introduction and Technical Context

The contemporary development of industrial infrastructures, cold-storage logistics centers, and hospital complexes demands highly efficient, safe, and durable thermal envelopes. Sandwich panels with high-density insulating cores represent the optimal engineering solution to meet these requirements.

Selecting the appropriate core material —whether Polyurethane (PUR), Polyisocyanurate (PIR), or Mineral Wool (MW)— depends on a precise balance between the following design criteria:

- **Thermal Resistance:** Minimizing energy losses or heat transfer by conduction.
- **Fire Safety:** Reaction to direct flame, flame spread rate, and toxic smoke generation.
- **Acoustic Insulation:** Air noise attenuation and sound wave absorption.
- **Mechanical Stability and Weight:** Structural load-bearing capacity relative to the span support of the panels.

2 Core Material Analysis

2.1 Polyurethane (PUR)

Polyurethane is obtained through an exothermic reaction between a diisocyanate and a polyol in the presence of blowing agents. It is a closed-cell material ($> 90\%$) optimized for exceptionally low thermal conductivity.

Key Properties:

- **Maximum thermal efficiency:** Delivers outstanding thermal resistance with thinner panel profiles.
- **Moisture resistance:** Its closed-cell structure prevents water infiltration and internal condensation.
- **Adhesion strength:** Excellent direct mechanical bonding to the steel skins of the panel.

2.2 Polyisocyanurate (PIR)

Polyisocyanurate is a direct chemical evolution of PUR. By modifying the stoichiometric ratio (isocyanate index) and introducing specific catalysts, the excess isocyanate reacts with itself to form trimerized isocyanurate rings. These rings feature significantly stronger and more thermally stable chemical bonds.

Key Properties:

- **Protective Carbonization:** When exposed to direct flame, PIR does not melt instantly. Instead, it forms a stable carbonaceous char layer on the surface that blocks heat transfer and delays structural collapse.
- **Low smoke emission:** Achieves Euroclass $s1$ or $s2$ rating in most commercial configurations.
- **Dimensional stability:** Retains its geometric properties across a broader operating temperature range.

2.3 Mineral Wool (MW / Rockwool)

Mineral Wool is manufactured by melting basaltic rock at temperatures exceeding $1500°$C, spinning it into multidirectional fibers, and compressing them with synthetic resin binders. Its open, fibrous structure differs drastically from plastic foams.

Key Properties:

- **Absolute non-combustibility:** Classified as Euroclass A1/A2. It withstands temperatures above $1000°$C without melting.
- **Premium acoustic damping:** The fibrous morphology effectively absorbs sound wave energy.
- **High density:** Roughly three times heavier than PIR/PUR, requiring a proper structural assessment of the support frame.

3 Technical Specifications and Performance Matrix

The following table summarizes the standardized quantitative and qualitative values for each core under European and international standards.

Table 1: Comparison of Insulating Core Technical Properties

Property	**PUR (Polyurethane)**	**PIR (Polyisocyanurate)**	**Mineral Wool (MW)**
Conductivity (λ)	~ 0.022 W/m · K	$\sim 0.020-0.022$ W/m·K	$\sim 0.035-0.040$ W/m·K
Reaction to Fire	Class C / D	Class B-s1, d0 / B-s2, d0	Class A1 / A2 (Non-combustible)
Density Range	$35-40$ kg/m^3	$38-42$ kg/m^3	$100-120$ kg/m^3
Acoustic Performance	Minimal	Standard	High ($R_w \sim 30-35$ dB)
Moisture Resistance	Excellent (Closed cell)	Excellent (Closed cell)	Hydrophobic (Open fibers)
Typical Application	Cold storage rooms	Logistics warehouses, offices	Chemical plants, hospitals

Note: Values are indicative and may vary depending on specific chemical formulations, manufacturer certifications, and metal sheet thicknesses.

4 Advanced Fire Safety and Certifications

Fire safety in commercial and industrial buildings is a critical factor for design compliance and financial insurability. The performance of sandwich panels is evaluated under strict European and global testing protocols.

4.1 The Euroclass System (EN 13501-1)

The harmonized European testing standard EN 13501-1 evaluates three fire risk parameters:

1. **Contribution to fire (Classes A1 to F):** Evaluates whether the material fuels the flames.
2. **Smoke opacity (Classes s1, s2, s3):** Where $s1$ denotes minimal visual interference from smoke.
3. **Flaming droplets (Classes d0, d1, d2):** Where $d0$ guarantees the complete absence of falling droplets that could spread secondary fires.

4.2 The PIR Chemical Mechanism: Protective Carbonization

Unlike thermoplastics (such as expanded polystyrene) or standard polyurethanes (PUR), which undergo thermal melting, shrinkage, and rapid disintegration under direct fire, PIR foam undergoes a physical-chemical process called **carbonization**.

When subjected to high temperatures, heat preferentially breaks down less stable chemical bonds, initiating the instant formation of a rigid, carbonaceous crust on the exposed surface.

Table 2: Euroclass Reaction to Fire Classification

Euroclass	Typical Material	Smoke Opacity (s)	Droplets (d)
A1	Pure Mineral Wool, Concrete	–	–
A2	Bonded Mineral Wool	s1	d0
B	High-Performance PIR	s1 / s2	d0
C	Optimized PUR	s2	d0
D / E	Standard Commercial PUR	s3	d2

This protective char layer has very low thermal diffusivity, blocking heat propagation deeper into the stable core and maintaining the structural integrity of the panel for periods up to 60 minutes.

4.3 FM Global (Factory Mutual) Approval

Certification under the **FM 4880** standard is recognized as the most demanding international test protocol for insulated panel systems. It evaluates the combustibility of panels at full scale in corner and wall configurations without requiring sprinkler systems. Specifying ”FM Approved” PIR cores substantially reduces insurance premiums and guarantees the resilience of industrial facilities against major fire disasters.

5 Thermal Performance and Transmittance (U-Value)

Thermal transmittance (U, measured in W/m$^2 \cdot$ K) represents the rate of heat transfer through a structure per unit area, induced by a temperature difference of one kelvin between the two environments separated by the panel.

5.1 Calculation Methodology according to EN ISO 6946

For a sandwich panel featuring a single homogeneous insulating core layer of thickness d, the mathematical formula for thermal transmittance is:

$$U = \frac{1}{R_{si} + \frac{d}{\lambda} + R_{se}}$$

Where:

- U: Total thermal transmittance (W/m$^2 \cdot$ K).
- d: Nominal thickness of the insulating material expressed in meters (m).
- λ: Thermal conductivity of the core (W/m $\cdot$ K).
- R_{si}: Interior surface thermal resistance (m$^2 \cdot$ K/W), typically standardized at 0.13 m$^2 \cdot$ K/W for horizontal heat flows.
- R_{se}: Exterior surface thermal resistance (m$^2 \cdot$ K/W), typically standardized at 0.04 m$^2 \cdot$ K/W.

5.2 Parametric Analysis: Thickness vs. Transmittance (U)

By using average standard thermal conductivity values ($\lambda_{\text{PIR}} = 0.021$, $\lambda_{\text{PUR}} = 0.022$, $\lambda_{\text{MW}} = 0.037$ W/m $\cdot$ K), the calculated transmittance values across commercial thicknesses are:

Table 3: Calculated U-Values (W/m$^2 \cdot$ K) by Thickness and Core Type

Panel Thickness (mm)	**PIR (U)**	**PUR (U)**	**Mineral Wool (U)**
50	0.39	0.41	0.66
80	0.25	0.26	0.43
100	0.20	0.21	0.35
120	0.17	0.18	0.30
150	0.14	0.14	0.24
200	0.10	0.11	0.18

As shown by the quantitative results in Table 3, PIR delivers approximately 5% better thermal performance than standard PUR and roughly 40% higher volumetric insulation than Mineral Wool for any given thickness. This explains why PIR remains the preferred core when space optimization and structural lightness are key priorities.

6 Project Specification Guidelines

Based on the integrated engineering analysis presented in this catalog, the following core specification guidelines are recommended:

1. **Specify PIR for High Efficiency and Standard Safety:** Recommended for 80% of modern industrial designs (logistics warehouses, retail centers, office spaces). PIR provides the highest thermal return on investment by significantly reducing HVAC electricity consumption while complying with strict building fire codes.
2. **Specify Mineral Wool for High-Risk Zones:** Imperative for firewall partitions, boiler rooms, chemical facilities, industrial kitchens, and hospitals where absolute structural non-combustibility (Class A1/A2) and acoustic damping are mandatory regulatory requirements.
3. **Specify PUR for Low-Height Cold Storage:** Suitable for controlled environments with strict thermal requirements, where high fire ratings are not a driving design constraint and lower initial material procurement costs are preferred.